LESSON 317 (1.5 DUAL 1.3 INSTRUMENT)

Lesson Objective
During this flight lesson, the student will increase proficiency by practicing VOR and GPS holding procedures.

Briefing
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
- Student questions.

Lesson Content (Ground)
Go/no-go Decision
- The student must make a go/no-go decision depending on:
  - Personal minimums.
  - FIT minimums.
  - Aircraft.
  - Airspace.
- Review the students planning with a focus on:
  - Weather, weight and balance, performance, flight planning, and fuel calculations.
  - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
  - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
  - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Lesson Content (Flight)
Aircraft Checklist Items

Emphasize timeliness to minimize ground time!

- Instrument Cockpit Check
  - <u>Student Tells, Student Does</u>.
    - Have the student read each item from the checklist out loud.
    - Discipline the student to set bugs, bearing pointers, and any auxiliary features.
  - The autopilot check can be conducted now:
    - Engage autopilot.
    - Test roll: left and right.
    - Test pitch: up and down (VS mode).
    - Ensure autopilot can be overridden.
    - Check the red disconnect button.

- Electric trim test:
    - Check the motion of the trim wheel.
    - Check the S-TEC trim annunciators.
    - Each half-switch should not move the trim wheel.
    - The disconnect switch should stop the wheel.
- Before taxi check:
  - <u>Student Tells, Student Does</u>.
    - Airspeed indicator (PFD and standby): zero.
    - Attitude indicator (PFD): pitch/bank deviations +/- 5 degrees.
    - Standby attitude indicator: standby battery check and adjust the miniature aircraft.
    - Altimeter (PFD and standby): +/- 75 feet of station elevation.
    - Vertical speed indicator (PFD): reads 0 or use current indication.
    - Trim coordinator: wings level or no vector when still; inclinometer centered when still; skids when turning.
    - HSI: matches compass and turns appropriately.
    - Magnetic compass: indicates heading properly, no bubbles, card present, and swings freely.

Departure

- Follow the tower's instructions for departure.
- The student must put the hood or foggles on when safe.
- Climb out:
  - <u>Student Does, Instructor Evaluates.</u>
    - Maintain V_Y.
    - Conduct the 1,000ft check.
    - Give vectors to depart Melbourne's airspace.
    - Level off at a given altitude.

VOR Holding Procedures

- Standard hold:
  - You can take the student to the VOR of your choice (MLB and TRV are the closest options).
- <u>Student Tells, Student Does.</u>
  - Give the student instructions to fly directly to the MLB VOR.
    - For example, track the 160-radial outbound, then track the 180 radial inbound while climbing above the Class D airspace to hold over the airspace.
  - The student can choose which method they would like to fly to the VOR:
    - Pressing OBS and flying the radial to the VOR. Ensure the ground track is on the radial.
    - Using the bearing pointer to fly to the VOR. This is the preferred method, which allows the CDI to be twisted to the inbound leg for the hold.

- Give the student a standard holding instruction (omit saying right turns).
    - *Hold south of the MLB VOR on the 180 radial, EFC (add 15 min to clock on MFD).*
- Ensure that the student repeats and understands the instructions.
- Ensure that the student can determine the correct entry procedure.
- Verify the completion of the 6 Ts (note that the order may change slightly depending on entry):
    - **T**urn - upon crossing fix **(flag flip)**, turn to the outbound heading.
        - This will be the given radial.
    - **T**alk - advise ATC established in the hold.
    - **T**ime - start the timer at **wings level** or **flag flip**, **whichever happens last.**
    - **T**wist - twist the CDI to the inbound course.
        - This will be the reciprocal of the given radial.
    - **T**rack - after one minute or the holding distance, turn inbound and track the inbound course.
    - **T**hrottle - adjust throttle for 90 kts.
- Ensure the student is making proper wind and time corrections.
    - Wind correction: once established inbound for the first time, note the difference between the current heading and the ground track on the PFD. This difference x 3 should be applied into the direction of the wind to the outbound heading.
    - Time correction: time each full inbound leg from wings level until reaching the fix. Note the difference between this time and one minute. Add or subtract this difference to the time of each outbound leg.
        - Note: do not time the first inbound leg as a part of the entry. This correction only applies to full inbound legs once established in the hold.
- While holding, ensure the student is continuing with the radial scan.
- Remind the student about the cone of confusion within 1 mile of the VOR and not make big corrections.

- Non-standard hold:
- <u>Student Tells, Student Does.</u>
    - Give the student vectors or a radial to intercept out of the previous holding pattern.
    - Once at least 3 NM away from the VOR, clear the student direct to the MLB VOR.
    - The student can choose which method they would like to fly to the VOR.
    - Give the student a non-standard holding instruction (omit saying right turns).
        - *Hold Northwest of the MLB VOR on the 320 radial, left turns, EFC (add 15 min to clock on MFD).*
    - Ensure that the student repeats and understands the instructions.
    - Verify the completion of the 6 Ts (note that the order may change slightly depending on entry):
        - **T**urn - upon crossing fix **(flag flip)**, turn to the outbound heading.
            - This will be the given radial.

    - **T**alk - advise ATC established in the hold.
    - **T**ime - start the timer at **wings level** or **flag flip**, **whichever happens last.**
    - **T**wist - twist the CDI to the inbound course.
      - This will be the reciprocal of the given radial.
    - **T**rack - after one minute or the holding distance, turn inbound and track the inbound course.
    - **T**hrottle - adjust throttle for 90 kts.
  - Ensure the student is making proper wind and time corrections.
    - Wind correction: once established inbound for the first time, note the difference between the current heading and the ground track on the PFD. This difference x 3 should be applied into the direction of the wind to the outbound heading.
    - Time correction: time each full inbound leg from wings level until reaching the fix. Note the difference between this time and one minute. Add or subtract this difference to the time of each outbound leg.
      - Note: do not time the first inbound leg as a part of the entry. This correction only applies to full inbound legs once established in the hold.
  - When holding, ensure the student is continuing with the radial scan.
  - Remind the student about the cone of confusion within 1 mile of the VOR and not make big corrections.

Give the student a vector out of the holding pattern once the student has demonstrated proficiency.

GPS Holding Procedures

- Standard hold:
- <u>Student Tells, Student Does.</u>
  - Clear the student direct to a GPS fix (e.g. VALKA, TPSTR, SMERE, MLBSO, etc).
  - If the student is unable to spell the fix, encourage them to ask for the spelling or use the nearest page to find the fix.
  - Give the student a standard holding instruction:
    - *Hold NE of the TPSTR intersection on the 040 bearing, EFC 15 mins.*
  - Ensure that the student repeats and understands the instructions.
  - Ensure that the student can determine the correct entry procedure.
  - Verify the completion of the 6 Ts (note that the order may change slightly depending on entry):
    - **T**urn - upon crossing fix, turn to the outbound heading.
      - This will be the given bearing.
    - **T**alk - advise ATC established in the hold.
    - **T**ime - start the timer at **wings level** or **flag flip**, **whichever happens last.**
    - **T**wist - **press OBS** and twist to the inbound course.
      - This will be the reciprocal of the given bearing.

        - **T**rack - after one minute or the holding distance, turn inbound and track the inbound course.
        - **T**hrottle - adjust throttle for 90 kts.
    - Ensure the student is making proper wind and time corrections.
        - Wind correction: once established inbound for the first time, note the difference between the current heading and the ground track on the PFD. This difference x 3 should be applied into the direction of the wind to the outbound heading.
        - Time correction: time each full inbound leg from wings level until reaching the fix. Note the difference between this time and one minute. Add or subtract this difference to the time of each outbound leg.
            - Note: do not time the first inbound leg as a part of the entry. This correction only applies to full inbound legs once established in the hold.
    - While holding, ensure the student is continuing with the radial scan.
- Non-standard hold:
- <u>Student Tells, Student Does.</u>
    - Clear the student direct to another GPS fix (e.g. VALKA, TPSTR, SMERE, MLBSO, etc.).
    - If the student is unable to spell the fix, encourage them to ask for the spelling or use the nearest page to find the fix.
    - Give the student a non-standard holding instruction:
        - *Hold SE of the VALKA intersection on the 120 bearing, left turns, EFC 15 mins.*
    - Ensure that the student repeats and understands the instructions.
    - Ensure that the student can determine the correct entry procedure.
    - Verify the completion of the 6 Ts (note that the order may change slightly depending on entry):
        - **T**urn - upon crossing fix, turn to the outbound heading.
            - This will be the given bearing.
        - **T**alk - advise ATC established in the hold.
        - **T**ime - start the timer at **wings level** or **flag flip**, **whichever happens last.**
        - **T**wist - **press OBS** and twist to the inbound course.
            - This will be the reciprocal of the given bearing.
        - **T**rack - after one minute or the holding distance, turn inbound and track the inbound course.
        - **T**hrottle - adjust throttle for 90 kts.
    - Ensure the student is making proper wind and time corrections.
        - Wind correction: once established inbound for the first time, note the difference between the current heading and the ground track on the PFD. This difference x 3 should be applied into the direction of the wind to the outbound heading.
        - Time correction: time each full inbound leg from wings level until reaching the fix. Note the difference between this time and one minute. Add or subtract this difference to the time of each outbound leg.

        - Note: do not time the first inbound leg as a part of the entry. This correction only applies to full inbound legs once established in the hold.
    - While holding, ensure the student is continuing with the radial scan.

Give the student vectors back to Melbourne when they have demonstrated proficiency.

Post-Brief

- Student Self Critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 318.
- Assign Homework:
    - Keep reviewing and practicing holding procedures.
    - Read the Instrument Flying Handbook:
        - Chapter 10 Section 13 through 22.

Completion Standards

At the completion of this lesson, the student shall:

1. Have the necessary skill and knowledge to determine the correct holding pattern entries and entry headings for VOR and GPS holding patterns – standard and non-standard.
2. Maintain the desired altitude within + 200 feet and airspeed within + 10 knots of the assigned.